Bhartṛhari's ~~Paradox~~ Limitation Theorem

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Abstract. In the Sambandha-samuddeśa of his *Vākyapadīya* (5th c. CE), Bhartṛhari argues that the relation between a word and its meaning cannot itself be signified. He then turns the claim on itself: to call the relation unsignifiable (*avācya*) is already to signify it. Herzberger and Herzberger named the passage "Bhartṛhari's paradox" and read it as a genuine paradox; Houben read it as a paradox that Bhartṛhari solves. We argue that it is a theorem, not a paradox, and that this reconciles the two readings. Using a set-theoretic version of Lawvere's fixed-point theorem, we show that the passage is a diagonal argument, of the same kind as Cantor's theorem and the Liar. Its conclusion is a limitation theorem: no term of a language can signify that language's own signification relation.

Keywords: Bhartṛhari · Vākyapadīya · diagonal argument · Lawvere's fixed-point theorem · paradox · history of logic

1 Introduction

Can the relation between a word and what it means itself be conveyed by a word? Bhartṛhari, writing on relation in the third book of the *Vākyapadīya*, answers no: no expression signifies the signification relation in its own nature (SS 4), and neither speaker nor hearer can reach it through words (SS 19). He then turns the signification relation on itself: if something is called unsignifiable, then it is thereby signified, by the word "unsignifiable" (SS 20) [1].

Two opposed readings of the self-reference have shaped the discussion. For Herzberger and Herzberger [1], who brought the passage to wide attention under the name "Bhartṛhari's paradox," the text records a genuine and stable paradox. They read it as a family of unnameability results, seven claims that the signification, denotation, and inherence relations each lie beyond naming, and take the whole to be one antinomy that recurs under every reformulation. For Houben [2], the passage instead offers a solution to the family of paradoxes to which the Liar belongs. The disagreement is about how to treat the paradox: for the Herzbergers the contradiction is real and survives restatement; for Houben the text already carries the means to defuse it. Two later readings side with Houben: Timalsina [5] holds the contradiction to be apparent (*virodhābhāsa*), and Oetke [6] rejects the alleged tension outright, holding that the passage defends the consistency of a semantic thesis and raises no truth-paradox at all.

We read the passage as a diagonal argument. A diagonal argument turns on three ingredients: an index for each object of the domain, a way for an object to stand for a property of objects, and an operation that inverts values. In Cantor's theorem the objects are the members of a set, an object stands for a subset through membership, and the inversion is complementation; in Turing's halting problem the objects are programs, a program stands for the halting behaviour it computes, and the inversion swaps halting for looping. In each, no object stands for the inverted diagonal property, and that is the limitative conclusion.

In 1969, Lawvere [7] gave an abstract generalisation covering many such arguments. Lawvere's fixed-point theorem states: if every property of objects is carried by some object, then every operation on values has a fixed point. Its contrapositive is the diagonal argument: negation on truth values has no fixed point, so some property of objects is carried by no object. In 2003, Yanofsky [8] recast this at the level of sets and functions, and derived the classical paradoxes, the incompleteness and undefinability theorems, and the recursion-theoretic fixed points as instances of one scheme.

The kārikās fit this scheme: words make properties of words available (SS 1), and negation switches values. Bhartṛhari's conclusion, that the relation cannot be signified, is then a limitation theorem. It is the signification analogue of the undefinability of truth. This reconciles the two readings. The Herzbergers are right that the structure is stable: it is a theorem. Houben is right that it is not a contradiction: a theorem is consistent, and the verses he reads as a solution defend the theorem against the charge of self-refutation. SS 20 is the evaluation step of the diagonal proof; SS 21 is the price of refusing that step; SS 22 is its strengthened form; and SS 25–28 weigh the standard responses to a limitation theorem, rejecting two by name and keeping the other two.

The correspondence (Sect. 4, Table 1) turns the Herzbergers' unsignifiability claims into corollaries of one theorem, without the axiom of regularity or the truth predicate that earlier derivations use.

2 The Text

The *Vākyapadīya* is a treatise on grammar and the philosophy of language in three books, attributed to the grammarian-philosopher Bhartṛhari (5th c. CE). The Sambandha-samuddeśa, "exposition on relation," is the third section of the third book; the part that concerns us is its opening segment, verses 1–28.[1]

The passage belongs to a standing debate over the limits of naming. The Naiyāyikas placed at the head of their works a saying the Herzbergers quote: "whatever is, is knowable and nameable" [1, p. 1]. Buddhist philosophers denied

[1] Sanskrit text follows Rau's edition as given in the GRETIL e-text [24], printed continuously with sandhi applied in both Devanagari and transliteration, and with *sambandha* for Rau's *saṃbandha*; verse 3.3.n is cited as SS n. In SS 27, Iyer and Houben read *naiva* for Rau's *nātra* [2, n. 14]. Translations of SS 1–25 follow Iyer [9], as quoted in [25], and those of SS 26–28 follow Oetke [6], with changes to SS 1, SS 19–22, and SS 25–27.

it. The Sambandha-samuddeśa takes up one case: can the relation that ties a word to its meaning itself be named? The chapter opens by fixing its subject matter:

> ज्ञानं प्रयोक्तुर्बाह्योऽर्थः स्वरूपं च प्रतीयते ।
> शब्दैरुच्चरितैस्तेषां सम्बन्धः समवस्थितः ॥
>
> *jñānaṃ prayokturbāhyo'rthaḥ svarūpaṃ ca pratīyate |*
> *śabdairuccaritaisteṣāṃ sambandhaḥ samavasthitaḥ* || (SS 1)
>
> From words that are uttered, the intention of the speaker, an external object and the form of the word itself are understood. Their relation is well-established.

By SS 1, what a word signifies may be the speaker's cognition (*jñāna*, rendered above as intention), an external object, or the word-form itself, and the signification relation (*sambandha*) is settled in advance of any particular utterance (*samavasthita*) [9, p. 76].

Two clarifications about this reading of SS 1. First, it grounds signification at the level of the word, whereas Bhartṛhari's fuller doctrine, the *sphoṭa*, locates meaning in the sentence taken as an undivided whole (*vākyasphoṭa*), the word-meaning (*padasphoṭa*) being reached by analysis (*apoddhāra*) for the grammarian's convenience [16,17,18]. The word-level reading is faithful to SS 1–28 and is all the argument uses; nothing below rests on the atomism it may suggest. Second, we call the relation "well-established" without settling the *nitya* or *kṛtaka* question, whether the bond between word and meaning is eternal (Mīmāṃsā school) or set by convention (Nyāya school) [28]: the diagonal limitation follows once the bond is stable at the moment of utterance, so it holds on either reading, as it would for a Saussurean or a Peircean sign.

SS 3 adds that the relation shows itself through the genitive: this word is expressive *of* that meaning, and thereby its "thatness" (*tattvam*) is indicated. Then comes the claim that the Herzbergers made famous:

> नाभिधानं स्वधर्मेण सम्बन्धस्यास्ति वाचकम् ।
> अत्यन्तपरतन्त्रत्वाद्रूपं नास्यापदिश्यते ॥
>
> *nābhidhānaṃ svadharmeṇa sambandhasyāsti vācakam |*
> *atyantaparatantratvādrūpaṃ nāsyāpadiśyate* || (SS 4)
>
> There is no verbal element (besides the genitive suffix) which denotes this relation in its essential property. Being extremely dependent, its own form is never cognised.

> प्राप्तिं तु समवायाख्यां वाच्यधर्मातिवर्तिनीम् ।
> प्रयोक्ता प्रतिपत्ता वा न शब्दैरनुगच्छति ॥
>
> *prāptiṃ tu samavāyākhyāṃ vācyadharmātivartinīm |*
> *prayoktā pratipattā vā na śabdairanugacchati* || (SS 19)
>
> Neither the speaker nor the hearer can reach through words that juxtaposition (*prāpti*), called inherence, which transcends the attributes of the expressed things.

So the relation between words and their objects cannot itself be reached through words; Helārāja's comment on SS 19 concludes that it "is truly unsignifiable (*avācya*)" [1, p. 5]. The next three kārikās take up the difficulty that calling the relation unsignifiable signifies it. Helārāja took them to be objections voiced on behalf of Naiyāyika critics, with the sequel as the answer [1, p. 10]. On either attribution the passage states a thesis and then defends it:

> अवाच्यमिति यद्वाच्यं तदवाच्यतया यदा ।
> वाच्यमित्यवसीयेत वाच्यमेव तदा भवेत् ॥
>
> *avācyamiti yadvācyaṃ tadavācyatayā yadā |*
> *vācyamityavasīyeta vācyameva tadā bhavet* || (SS 20)
>
> If what is said to be inexpressible is expressed as inexpressible, it is tantamount to concluding it as expressible, and thereby it would indeed become expressible.

> अथाप्यवाच्यमित्येवं न तद्वाच्यं प्रतीयते ।
> विवक्षितास्य यावस्था सैव नाध्यवसीयते ॥
>
> *athāpyavācyamityevaṃ na tadvācyaṃ pratīyate |*
> *vivakṣitāsya yāvasthā saiva nādhyavasīyate* || (SS 21)
>
> Even then (i.e., just because we describe by the word 'inexpressible') it does not become expressible. It would be cognised only as inexpressible. [If we do not even use this phrase], then the condition that is meant to be conveyed would not be comprehended.

> तथान्यथा सर्वथा च यस्यावाच्यत्वमुच्यते ।
> तत्रापि नैव सावस्था तैः शब्दैः प्रतिषिध्यते ॥
>
> *tathānyathā sarvathā ca yasyāvācyatvamucyate |*
> *tatrāpi naiva sāvasthā taiḥ śabdaiḥ pratiṣidhyate* || (SS 22)
>
> Either this way, some other way, or in all ways, when an entity is described as inexpressible, even then those words would not deny that particular condition.

SS 23–24 turn to cognitions that take other cognitions as objects (doubt about a doubt, ascertainment of an ascertainment) and deny that a cognition can operate on itself while retaining its own character (*svadharme nāvatiṣṭhate*). SS 25 then gives a version of the Liar, or possibly, since *mithyā* can mean false or unreal, a declaration that is made from the perspective of Advaita Vedānta:

> सर्वं मिथ्या ब्रवीमीति नैतद्वाक्यं विवक्ष्यते ।
> तस्य मिथ्याभिधाने हि प्रक्रान्तोऽर्थो न गम्यते ॥
>
> *sarvaṃ mithyā bravīmīti naitadvākyaṃ vivakṣyate |*
> *tasya mithyābhidhāne hi prakrānto'rtho na gamyate* || (SS 25)
>
> Just because I utter 'all I say is false', that sentence is not intended to be included. If that (i.e., what it says) is also false, the point in question would not be conveyed.

Three closing verses then provide more examples to drive in the point that has been made:

> न च वाचकरूपेण प्रवृत्तस्यास्ति वाच्यता ।
> प्रतिपाद्यं न तत्तत्र येनान्यत्प्रतिपाद्यते ॥
>
> *na ca vācakarūpeṇa pravṛttasyāsti vācyatā |*
> *pratipādyaṃ na tattatra yenānyatpratipādyate* || (SS 26)
>
> What is operative as a signifier is not something to be signified; that by which something else is conveyed is not, in the same context, to be conveyed.

> असाधिका प्रतिज्ञेति नेयमेवाभिधीयते ।
> यथा तथास्य धर्मोऽपि नात्र कश्चित्प्रतीयते ॥
>
> *asādhikā pratijñeti neyamevābhidhīyate |*
> *yathā tathāsya dharmo'pi nātra kaścitpratīyate* || (SS 27)
>
> Just as, [in a *Nyāya-prayoga*,] 'the proposition (*pratijñā*) [alone] does not prove', this very proposition is not designated, so too no characteristic of it is ascertained there.

> व्यापारस्यापरो यस्मान्न व्यापारोऽस्ति कश्चन ।
> विरोधमनवस्थां वा तस्मात्सर्वत्र नाश्रयेत् ॥
>
> *vyāpārasyāparo yasmānna vyāpāro'sti kaścana |*
> *virodhamanavasthāṃ vā tasmātsarvatra nāśrayet* || (SS 28)
>
> Because there is no further operation of an operation, one should not resort to a contradiction (*virodha*) or an infinite regress (*anavasthā*) in any case.

One ambiguity decides what a formal reconstruction has to capture. Parsons [4, p. 532] distinguishes three questions: whether a *noun* such as "signification" can signify signification, which was the Herzbergers' question; whether a *verb* such as "signifies" can, which was his; and whether an *adjective* such as "unsignifiable" can apply to signification, which is what the verses, framed around *avācya*, ask. This last, he writes, neither he nor the Herzbergers addressed. The reconstruction below runs on the adjective and yields the noun and verb versions as variations.

3 The Diagonal Scheme

This section uses Yanofsky's set-theoretic form of Lawvere's theorem, with $Y = \{0, 1\}$ in the instance we need. Given a function $f\colon T \times T \to Y$, say that $g\colon T \to Y$ is *representable by* f if there is $t_0 \in T$ with $\forall x.\, f(x, t_0) = g(x)$. Think of T as a stock of expressions, Y as a set of values, each map $x \mapsto f(x, t_0)$ as the property that the index t_0 makes available, and representability as being available under some expression. We call f the *evaluator*.

Theorem 1 (Cantor; [8, Thm. 1]). *Let $\alpha\colon Y \to Y$ have no fixed point. Then for every set T and every $f\colon T \times T \to Y$, the function $g\colon T \to Y$ defined by $g(t) = \alpha(f(t,t))$ is not representable by f.*

Proof. If $\forall x.\, f(x, t_0) = g(x)$, evaluate at t_0: $f(t_0, t_0) = g(t_0) = \alpha(f(t_0, t_0))$, a fixed point of α. □

Yanofsky [8], following Lawvere [7], obtains the classical limitative results by choosing T, Y, f, and α: Grelling's paradox with T the adjectives and $f(a_1, a_2) = 1$ iff a_2 describes a_1; the Liar over sentences; Russell over sets; Tarski and Gödel over formulas.

The constructed g is definable from f by two operations, diagonalisation and composition with α. So if the indexing system is closed under those operations and f itself is available as an index, the unrepresentable g would become representable, a contradiction. Hence the evaluator f is itself what cannot be represented: an indexing system closed under diagonalisation, with a fixed-point-free switch, cannot index its own evaluator.

A three-valued instance answers the proposal that paradoxical sentences be set aside as meaningless: take the values true, meaningless, and false, and let α send true to false and the other two to true. This is the strengthened Liar [8, Sect. 2]. The switch again has no fixed point, so the diagonal argument runs over the enlarged value set. Adding more values does not remove the limitation.

Remark 1. Nothing here depends on the category of sets. Lawvere's theorem holds in any Cartesian closed category [7, Thm. 1.1], and his Theorem 2.2 needs only finite products, which a language category provides. In the Lindenbaum category of a theory, where constants $1 \to A$ name formulas, his Corollary 3.1 reads: if satisfaction is definable in the theory, the theory is inconsistent. The contrapositive is the undefinability of satisfaction; Lawvere's Theorem 3.2 derives Tarski's theorem on truth from it when substitution is definable. The claim we attribute to Bhartṛhari has this form, with signification in place of satisfaction.

4 The Correspondence

Fix a set T of terms. What a term can mean (its *artha*) is, for Bhartṛhari, a cognition, an external object, a word-form (SS 1), a universal, or a relation (SS 3). A property of terms is a function $T \to \{0,1\}$. Since SS 1 declares the relation well-established, which arthas a term signifies is settled prior to any utterance. A term *signifies* a property $P\colon T \to \{0,1\}$ when P is among its arthas; the relation f below can itself be an artha, since relations are (SS 3). A term may signify several properties of terms. So that f has one value at each pair, we treat a term that signifies several properties of terms as several elements of T, one for each. Each $u \in T$ then signifies at most one property of terms, written P_u when it exists. Define the evaluator

$$f\colon T \times T \to \{0,1\}, \qquad f(t,u) = \begin{cases} P_u(t) & \text{if } P_u \text{ exists,} \\ 0 & \text{otherwise,} \end{cases}$$

so that if u signifies P then $\forall x.\, f(x,u) = P(x)$. Signifiability entails representability by f, the direction the proof uses. The proof needs one premise beyond the setup.

Assumption 1 (Closure). *If the relation f is signified by some term, then so is any property obtained from f by diagonalisation and value-switch, in particular $g(t) = \alpha(f(t,t))$ with α the swap on $\{0,1\}$.*

Closure is our reconstruction; the text does not state it as a premise. It has two parts. Diagonalisation evaluates a term on itself, and Bhartṛhari allows a word to take its own form as an object: by SS 1 the word-form (*svarūpa*) is among what a word conveys. Navya-Nyāya has a device for composing a diagonal with negation. To name, without a free variable, the property of an x that is co-located with its own absence, it qualifies x by an absence through the double relation *svapratiyogitva-svasamānādhikaraṇya*: the reflexive *sva* binds two argument places of one relation to the same item, and the absence supplies the switch [27, pp. 85–86, n. 10]. That is the form of g. The kārikās use the relative–correlative pair for the same purpose. SS 26 binds signifier and signified to one item with *yena* … *tat*, and Helārāja's gloss states the schema without the context: *yat pratipādakaṃ na tat pratipādyam*, "what conveys is not what is conveyed" [26, p. 140]. The pronouns *yat* and *tat* fill both argument places of the signification relation with one item, and the negation applies to the result, as in $g(t) = \alpha(f(t,t))$. A speaker who can talk about f and about negation can form g, and word-meanings and their relations are themselves reached by abstraction and analysis (*apoddhāra*) from sentence-wholes [16].

Parsons [4] observed that the Herzbergers' argument needed supplements at two points, the privileging of regularity among the ways of blocking the paradoxes and the attribution to Bhartṛhari of set-extensions for relations; Assumption 1 replaces both.

Helārāja's commentary could be read against Closure. On SS 23 he says that the word *avācya*, excluding itself (*ātmānaṃ varjayan*), makes its object expressed, as a doubt makes its object doubtful without doubting itself. On SS 24 he says that the word, while conveying, does not deny expressibility of itself, because an act cannot be directed at its own self (*svātmani kriyāvirodhāt*); on SS 26, that what is engaged in conveying something else has, at that very time (*tadaiva*), no operation on itself [26, pp. 139–140]. Taken as a ban on diagonalisation, this would deny Closure. The ban is tied to the act and the occasion, as Houben stresses for *tatra* in SS 26 [2, p. 393]. It does not prevent forming g or signifying it by another term; it restricts the evaluation of an utterance on itself, the step SS 20 takes, and that is the scope restriction of Sect. 5. Houben leaves open whether the principle extends to a word revealing itself [2, p. 393, n. 15]; on this reading it need not.

Lemma 1. *The property $g\colon T \to \{0,1\}$, $g(t) = \alpha(f(t,t))$, is signified by no term.*

Proof. Theorem 1 with $Y = \{0,1\}$ and α the swap, which has no fixed point. □

Lemma 1 is Cantor's theorem in this setting. It holds of any f and uses no premise about signification; Closure is what turns it into the theorem below.

Theorem 2. *Under Assumption 1, no term signifies the relation f.*

Proof. If a term signified f, Assumption 1 would give a term signifying g, contradicting Lemma 1. □

This is SS 4 and SS 19 as the text states them. What no term signifies is the relation itself (*sambandha*), not some particular artha. The unsignifiability (*avācyatā*) is a result of the argument, not an assumption, and it concerns only the nameability of the relation, not any wider claim about language and thought. Each of the Herzbergers' seven claims R1–R7 [1] pairs one of three relations (signification, denoting, inherence) with one of three modes (signifying, denoting, naming), and each is the theorem rerun with the evaluator and the mode of indexing varied. Their observation that the paradox is "a theme with many variations" is the scheme run with f, α, and Y varied.

Since g is the extension of the adjective *avācya* over terms, the argument answers Bhartṛhari's own formulation, the one Parsons left open (Sect. 2). A noun naming f or a verb expressing it would equally make the evaluator available as an index, so those versions follow.

Our argument differs from earlier ones in two ways. The Herzbergers derived R1–R7 from the principle that no relation can be one of its own relata, resting on the axiom of regularity, and their note acknowledges non-regular systems such as Quine's NF, where a derivation from regularity is unavailable [1, n. 19]; Lemma 1 uses no foundation axiom and holds in such systems as well. Parsons recast their argument as a semantic one, defining a truth predicate from signification and rederiving the Liar [4]; our argument uses signification directly and does not go through a truth predicate.

The scheme also answers the objection, pressed by Parsons, that SS 3 reinstates the contradiction: the thatness of the relation is signified, so the relation is signified after all. In the model the two verses concern different levels. Each instance, that this word signifies that meaning, is a single value $f(t, u)$ and is expressible; that is SS 3, and it matches the Herzbergers' gloss of thatness as the relation's concrete manifestations [1, p. 7]. What no term provides is an index whose column is the whole evaluator; that is SS 4. Theorem 2 denies the second and leaves the first intact. The same distinction lets this paper write down f and g: we describe f from outside the language, and the theorem denies only that some term indexes it. Helārāja draws it on SS 4: the relation, once extracted and named by the word *sambandha*, is presented as a relatum (*sambandhirūpatayā abhidhīyamānaḥ*) and so "cannot be touched directly" (*na sākṣāt spraṣṭuṃ śakyate*) [26, p. 128].

Suppose, with SS 20, that the unsignifiable is signified as unsignifiable: a term t_0 signifies g. Evaluate at t_0, as the proof of Theorem 1 does: $f(t_0, t_0) = g(t_0) = \alpha(f(t_0, t_0))$. The value flips, and it "would indeed become expressible" (*vācyameva tadā bhavet*). SS 21 answers that describing it by the word does not make it expressible: it is cognised only as inexpressible. The verse's second

Table 1. The correspondence.

Diagonal scheme	*Vākyapadīya*
evaluator f fixed, prior to use	SS 1 (*sambandhaḥ samavasthitaḥ*)
evaluator not an index of itself	SS 4 (*svadharmeṇa*, *atyantaparatantratvāt*)
pointwise values expressible	SS 3 (*tattvam* via the genitive)
definability fails at f	SS 19; Herzbergers' R1–R7
evaluation step at t_0	SS 20 (*vācyameva tadā bhavet*)
g unrepresentable; regress	SS 21 (the condition is not conveyed)
rerun over refined values	SS 22 (*tathānyathā sarvathā ca*)
no self-application of operations	SS 23–24, SS 26–27
scope restriction for the Liar token	SS 25 (*naitadvākyaṃ vivakṣyate*)
no contradiction, no hierarchy	SS 28 (*virodhamanavasthāṃ vā*)

half is the other horn. Refuse the representing term, and "the condition that is meant to be conveyed would not be comprehended": if no term signifies g, no term conveys the condition g describes. Houben reads the verse as a regress: the unsignifiability of X is itself unsignifiable, and so on without end [2, pp. 387–388]. On either reading the refusal repeats: each level's unsignifiability passes to the next, which is the ascent through a hierarchy of metalanguages. Houben calls the two horns together "as strong as a strengthened paradox" [2, p. 388].

SS 22 is contested. It is a continuation of the objection for Helārāja and the start of the solution for Houben: words that signify X as unsignifiable do not thereby deny that X is unsignifiable [2, pp. 388–389]. On the objection reading the verse is the strengthened form: allege the unsignifiability "this way, some other way, or in all ways," and still "those words would not deny that particular condition"; refining the value set, as in the three-valued strengthened Liar, produces no fixed point for the switch, so the limitation survives every repartition of values. Houben observes that the verse covers "even stronger cases" [2, p. 389]. On the solution reading the verse distinguishes attributing *avācyatā* to X from signifying X by its own nature (*svadharmeṇa*, SS 4), which is the distinction drawn above for SS 3 and SS 4. The scheme accommodates both, and on either the verse registers that no reclassification of values dissolves the limitation.

Table 1 assembles the reading. One row deserves a caution. SS 4 grounds the relation's unnameability in its extreme dependence (*atyantaparatantratvāt*), and dependence is not self-membership; the verse admits readings, Oetke's regress reading among them [6], on which its ground differs from ours. We claim that the conclusion of SS 4 is the conclusion of Theorem 2, and that the surrounding verses supply the premises and the proof-steps of the diagonal route to it.

5 The Exits

A theorem, unlike a paradox, needs no solution, but it forces a choice among responses. The modern responses to it cluster into four exits:

- accept the contradiction
- stratify the language so that each evaluator is indexed one level up
- keep one language but let the evaluation be partial
- restrict, occasion by occasion, what a self-directed utterance quantifies over

Modern logic has taken all four, as dialetheism [21], the Tarski hierarchy [20], Kripke's partial truth [19], and contextualist accounts of the Liar. The closing kārikās consider the same four exits and choose.

Two are refused by name. SS 28 instructs: nowhere accept a contradiction, nowhere accept an infinite regress. *Virodha* is the dialetheist exit. *Anavasthā* is the hierarchy, and the identification is not a modern imposition: the regress that SS 21 threatens, unsignifiability$_1$ being unsignifiable$_2$, and so on, is what a tower of metalanguages accepts as a standing condition. Nor is the hierarchical exit a straw man here. This is Parsons's defence of Bhartṛhari: a signification relation Z which could be signified, except that signifying it creates a new signification relation for the extended language, and so on [4, p. 531]. Davis, in the earliest formal study of the passage, argued on textual grounds that language hierarchies cannot represent Bhartṛhari's position [12]. Coward gives the reason in discussing origins: the question is asked within language, and "we cannot get outside of language" to ask it from elsewhere [15, p. 6]. A universal language has no larger language to move up to.

The text takes the remaining two exits. The first is Cantor's: keep the language universal, keep consistency, and let definability fail at the evaluator. That is where Theorem 2 put the failure, and where Bhartṛhari put it two verses into the chapter (SS 4). The cost is expressive: SS 21 concedes that the intended condition then cannot be conveyed. Kripke's construction pays the same price: the status of ungrounded sentences cannot be stated inside the object language. The same trade-off holds in both: a language that is both universal and consistent must be expressively incomplete.

The second exit is to restrict the scope of the sentence. SS 25 says the sentence "is not intended to be included": the universal quantifier of "all I say is false" does not, in the intended use (*vivakṣā*), include the present utterance, else the speaker's point (*prakrānto'rthaḥ*) would be frustrated. The restriction is the same whether the verse is a Liar or an Advaita declaration that all is unreal. Houben describes this in modern terms: the universally quantified Liar, solved "by disallowing quantification over itself" on the basis of a factor Western accounts usually bracket off, the intention of the speaker [2, p. 392]. This is not the cassationism of some Latin medievals, on which the Liar utterer says nothing [22]: here the sentence is meaningful and its point gets made. It restricts the scope of the sentence, the move that contextualist accounts of the Liar develop systematically, and SS 26 ties the exclusion to the occasion: what proceeds as signifier is not, *tatra*, in that context, the signified [2, p. 393]. Nakamura, in the

earliest modern discussion, saw the verse's role in Bhartṛhari's programme: it defends sweeping doctrinal claims, his own "all inference is unreliable" included, against the self-refutation charge that the Naiyāyikas pressed on Nāgārjuna's "all things are void" [14, pp. 297–298]. The defence works by restricting scope, and it rests, through SS 23–24 and SS 26–27, on the principle that no operation applies to itself.

The two later readings also fit here. Oetke reached a neighbouring result by philological means alone: SS 19 states a consistent general thesis, SS 20–21 poses a dilemma about its quantifier, unrestricted and self-defeating or restricted and uncommunicative, and the sequel dissolves the dilemma [6]. His dilemma is the pair of horns every limitation theorem faces, and his distinction between representing and non-representing reference tracks ours between representation by an index and definition from outside. That Oetke reaches the same verdict from the Sanskrit alone is independent evidence that the diagonal structure is in the text, not imposed on it. Priest's inclosure schema [21], under which limits-of-expression paradoxes standardly license dialetheia, applies to this passage as well; what blocks the dialetheic reading is SS 28 itself, which forbids it directly.

6 Related Work and Conclusion

Besides the readings discussed above, Ganguli [10] compared Bhartṛhari's solution to Russell's, and Biardeau [11] discussed the passage; Matilal [16] concurred with the Herzbergers; Athale [25] surveys the positions and calls for a set-theoretic treatment; Coward [15] and Berger [23] read the *Vākyapadīya*'s limits of language through Derrida and through Pratyabhijñā realism. On the formal side the scheme is Lawvere's [7] and the set-level presentation Yanofsky's [8]. We are not aware of an earlier reading of the passage as a diagonal argument.

The Herzbergers' paradox and Houben's solution are two parts of one result: the limitation theorem and its defence against self-refutation. The one premise the kārikās do not state is Closure. SS 1, the Navya-Nyāya *sva*-device, and the *yat*/*tat* construction support it, and Helārāja's ban on self-application, read as tied to the occasion, restricts scope rather than denying it.

Two tasks remain open. We have used Helārāja's remarks on SS 4, SS 20, and SS 23–26; Houben's translation of the whole commentary [3] should be checked against Assumption 1. And the partial-signification reading of SS 21 invites a construction: a Kripke-style least fixed point for a signification jump, in which the status of *avācya* sentences, grounded, ungrounded, or paradoxical, can be computed rather than disputed. We intend to pursue it, starting from Davis's reformulation [13], which was built from this passage.

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